

# Does In-Distribution Temperature Scaling Stay Calibrated Under Shift? A Small Study on Digits

Ansatz AI research agent

Library seed corpus, commissioned by Mihir Chauhan (github.com/mihir-chauhan)

Topic, experiments, analysis and text generated by the agent

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/calibration-under-shift-digits>

**Abstract**—We ask whether temperature scaling (TS) fitted on in-distribution validation data stays calibrated under distribution shift, on scikit-learn digits with rotation (0–60°) and Gaussian-noise shift of increasing intensity. We reimplement and compare a single MLP, TS, MC dropout and a 5-member deep ensemble over 11 shift conditions and 5 seeds, reporting accuracy, ECE, NLL and Brier score. In distribution all systems are well calibrated and the TS gain is not statistically supported with 5 seeds. Under shift, ID-fitted TS does not remain calibrated: its ECE grows about 50-fold at 60° rotation and about 39-fold at noise  $\sigma = 1$ , although it still lowers NLL relative to the uncalibrated MLP at every shifted condition. An oracle ablation fitting the temperature on shift-matched validation data brings ECE back to at most 0.092, but at strong shift only by pushing predictions towards uniform (NLL close to  $\ln 10$  at 60°). Against our pre-registered expectation, MC dropout had lower mean ECE, NLL and Brier than the deep ensemble in all 11 conditions (paired differences significant at  $\alpha = 0.05$ , uncorrected, for NLL in all 11 conditions, ECE in 9 and Brier in 7; in distribution the ECE difference is within seed noise); an exploratory decomposition suggests this is partly an effect of dropout regularisation during training. The study is small (one dataset, one MLP, 5 seeds, untuned reimplemented baselines) and the conclusions are limited to that scale.

**Index Terms**—calibration, distribution shift, temperature scaling, deep ensembles, MC dropout

## I. Introduction

A classifier deployed in practice sees inputs that differ from its training data, and the usefulness of its probabilities then depends on whether they remain honest. Post-hoc temperature scaling is the cheapest calibration fix [1], but it is fitted on data that look like the training distribution. Whether the fitted temperature transfers to shifted inputs is an empirical question that large-scale benchmarks have studied on image corruptions [2] (building on the corruption benchmark of Hendrycks and Dietterich [3]), with the conclusion that in-domain calibration does not carry over reliably [2], [4].

We ask the same question at a scale where every cell of the experiment can be run in seconds on a CPU and repeated over five seeds: scikit-learn digits (8×8 images),

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an MLP, and two shift families with a graded intensity, rotation and additive Gaussian noise. We compare a single MLP, TS fitted on in-distribution validation data, MC dropout [5] and a five-member deep ensemble [6], measured by accuracy, expected calibration error (ECE), negative log-likelihood (NLL) and Brier score. The question that motivated the study is: does temperature scaling fitted in-distribution stay calibrated under shift? We register five hypotheses in advance (Section IV), and add an oracle-temperature ablation and two sensitivity sweeps to probe the mechanism.

Contributions are modest by design: (i) a small-scale reproduction of the shift-calibration comparison on a different data modality and two shift families; (ii) a controlled ablation that separates “the temperature is wrong under shift” from “a scalar cannot fix shifted predictions”; (iii) all code, run logs and tables in the repository. Everything is reimplemented here, nothing is tuned, and the conclusions are limited to this dataset, architecture and shift construction.

## II. Related Work

Guo et al. [1] showed that modern networks are overconfident and introduced temperature scaling (TS), a one-parameter post-hoc rescaling of the logits fitted on held-out in-distribution data, together with the ECE/NLL evaluation protocol we use. Ovadia et al. [2] then asked how predictive uncertainty behaves under dataset shift, comparing a plain network, TS, MC dropout, deep ensembles and other methods over graded shift intensities on image corruptions and other data; their headline finding was that quality degrades with shift and that ensembles were the most robust of the methods they compared. Our setup is a small-scale analogue of that protocol, with corruption-style shift intensities in the spirit of [3]. Tomani et al. [4] argue that post-hoc calibration fitted in-domain breaks under domain drift and propose to fit it on perturbed calibration data; our oracle-temperature ablation is a cheap version of that idea, using validation data transformed exactly like the test condition.

Deep ensembles [6] average independently trained networks; MC dropout [5] keeps dropout active at test time and averages stochastic passes as an approximation to

Bayesian inference, a reading that Le Folgoc et al. [7] question. Uncertainty Baselines [8] provides tuned large-scale implementations of these and related methods; Laplace approximations [9], SWAG [10] and architecture-diverse ensembles [11] are further approaches that we do not evaluate. We do not propose a new method; the contribution is a controlled, fully reproducible CPU-scale comparison with a mechanism ablation, and its conclusions should be read at that scale.

### III. Method

Let  $f_\theta(x) \in \mathbb{R}^K$  be the logits of an MLP ( $K = 10$ ) and  $p_\theta(y|x) = \text{softmax}(f_\theta(x))$ . All networks are trained with cross-entropy,  $\mathcal{L}(\theta) = -\frac{1}{N} \sum_i \log p_\theta(y_i|x_i)$ .

Single MLP. Prediction  $\hat{p}(y|x) = p_\theta(y|x)$  with a single trained network.

Temperature scaling (reimplemented from [1]). A scalar  $T > 0$  rescales logits,  $\hat{p}_T(y|x) = \text{softmax}(f_\theta(x)/T)$ . With the network frozen,  $T = \arg \min_T -\frac{1}{N_v} \sum_{j \in \mathcal{V}} \log \hat{p}_T(y_j|x_j)$ , optimised over  $\log T$  by L-BFGS on an in-distribution validation split  $\mathcal{V}$ . The argmax, and hence accuracy, is unchanged. The same  $T$  is applied to every shifted test condition.

Oracle TS (ablation). Identical, but  $\mathcal{V}$  is replaced by the validation images transformed by the same shift  $s$  as the test condition,  $T_s = \arg \min_T \text{NLL}(\hat{p}_T; s(\mathcal{V}))$ . This uses information (the shift) that is unavailable in deployment and serves only to isolate the mechanism.

MC dropout (reimplemented from [5]). The network is trained with dropout (rate  $p$ ) after each hidden layer; at test time dropout stays active and  $\hat{p}(y|x) = \frac{1}{S} \sum_{s=1}^S p_{\theta, \epsilon_s}(y|x)$  with  $S = 20$  masks  $\epsilon_s$ .

Deep ensemble (reimplemented from [6]).  $M = 5$  networks differing in initialisation and minibatch order,  $\hat{p}(y|x) = \frac{1}{M} \sum_m p_{\theta_m}(y|x)$ . We use no adversarial training and no proper-scoring-rule variants beyond cross-entropy.

Metrics. With confidence  $c_i = \max_k \hat{p}(k|x_i)$ , prediction  $\hat{y}_i$  and bins  $B_b$  (15 equal-width bins on  $(0, 1]$ ),

$$\text{ECE} = \sum_b \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)|, \quad (1)$$

$$\text{NLL} = -\frac{1}{n} \sum_i \log \hat{p}(y_i|x_i), \quad (2)$$

$$\text{Brier} = \frac{1}{n} \sum_i \sum_k (\hat{p}(k|x_i) - \mathbb{I}[k = y_i])^2. \quad (3)$$

We also report accuracy and the mean confidence  $\bar{c}$ . NLL and Brier are proper scoring rules; ECE depends on binning and is reported alongside them, not instead of them.

Shift. For a test image  $x \in [0, 1]^{64}$ , rotation by  $\alpha$  degrees (bilinear interpolation, zero fill, no resize) gives  $R_\alpha x$ , and noise gives  $\text{clip}(x + \sigma \varepsilon, 0, 1)$  with  $\varepsilon \sim \mathcal{N}(0, I)$ ; the combined condition applies rotation, then noise. Rotation by  $\alpha = 0$  with no noise is the in-distribution condition.

### IV. Experimental Setup

Data and splits. scikit-learn digits (1797 images,  $8 \times 8$ , pixel values divided by 16). For each seed  $s \in \{0, 1, 2, 3, 4\}$  a stratified split gives 1078 training, 269 validation and 450 test images; the seed also determines network initialisation, minibatch order, dropout masks and noise. Shift is applied to test images only (and, for the oracle ablation, to the validation images), never to training data.

Shift conditions (“tasks”). Eleven conditions: in-distribution (r0); rotations of 10, 20, 30, 45, 60 degrees (r10–r60); Gaussian noise with  $\sigma \in \{0.25, 0.5, 0.75, 1.0\}$  (n0.25–n1.0); and rotation 30 plus noise 0.5 (r30+n0.5). The strongest conditions are not mild: on  $8 \times 8$  images accuracy falls to chance level for r45/r60 (Table II), where ECE is therefore essentially the mean confidence.

Systems. All reimplemented in PyTorch in method/run.py: MLP 64–128–128–10 with ReLU; Adam, learning rate  $10^{-3}$ , batch 64, 100 epochs, no weight decay, no early stopping. MLP+TS uses the MLP with  $T$  fitted on unshifted validation data; MC dropout trains the same architecture with dropout  $p = 0.2$  after each hidden layer and averages  $S = 20$  stochastic passes; the ensemble has  $M = 5$  MLPs. The single MLP is ensemble member 0. All systems share data, splits, optimiser and epochs, and are scored with the same metric code (15-bin ECE, NLL, Brier, accuracy, mean confidence). Network weights are cached per (architecture, seed) so that the rows of a seed share one trained network; training is deterministic.

Tuning budget and tests. None: every hyperparameter was fixed before the runs and not selected on test data (validation data are used only to fit  $T$ ). The sweeps in Section VI are reported, not used to choose settings. Hypotheses H1–H5 were registered in proposal.md before the main runs. H1, H4 and H5 are tested with paired  $t$ -tests over the five seeds at  $\alpha = 0.05$  (no correction for multiple comparisons, so they are descriptive); H2 and H3 are comparisons of means over seeds. The dropout decomposition (Section VI) was added after seeing the main results and is exploratory.

Compute. Apple-silicon CPU (shared machine), 2 threads, torch CPU. One run takes a few seconds; the 412 logged runs (main 220, oracle 55, dropout decomposition 55, ensemble-size sweep 45 and dropout-rate sweep 36 on three conditions and seeds 0–2, one sanity run) took about 40 minutes of summed run time (about 32 minutes wall-clock) under load from other jobs. Paired  $t$ -test  $p$ -values are computed by results/analysis.py (scipy), not by rh compare.

### V. Results

We go hypothesis by hypothesis. Per-seed standard deviations are in the generated tables (results/tables/main.md); with five seeds, all verdicts are descriptive.

TABLE I

ECE and NLL (mean over 5 seeds; lower is better, bold marks the lowest mean only, not a significant difference) per shift condition.  $\alpha$ : rotation by  $\alpha^\circ$ ;  $\text{n}\sigma$ : Gaussian noise; MCD: MC dropout; Ens: 5-member ensemble; TS: MLP with ID-fitted temperature.

| Shift    | ECE MLP | ECE TS | ECE MCD | ECE Ens | NLL MLP | NLL TS | NLL MCD | NLL Ens |
|----------|---------|--------|---------|---------|---------|--------|---------|---------|
| r0       | 0.018   | 0.016  | 0.016   | 0.017   | 0.117   | 0.104  | 0.083   | 0.111   |
| r10      | 0.104   | 0.070  | 0.042   | 0.078   | 0.745   | 0.614  | 0.491   | 0.645   |
| r20      | 0.226   | 0.161  | 0.149   | 0.208   | 1.954   | 1.450  | 1.370   | 1.687   |
| r30      | 0.457   | 0.388  | 0.361   | 0.415   | 4.747   | 3.365  | 3.098   | 3.987   |
| r45      | 0.816   | 0.785  | 0.752   | 0.803   | 13.431  | 9.490  | 8.639   | 12.257  |
| r60      | 0.827   | 0.803  | 0.792   | 0.820   | 15.266  | 11.101 | 10.442  | 13.874  |
| n0.25    | 0.105   | 0.067  | 0.043   | 0.083   | 0.763   | 0.587  | 0.479   | 0.667   |
| n0.5     | 0.392   | 0.335  | 0.280   | 0.343   | 4.109   | 2.903  | 2.402   | 3.467   |
| n0.75    | 0.584   | 0.524  | 0.474   | 0.522   | 7.746   | 5.468  | 4.525   | 6.597   |
| n1.0     | 0.674   | 0.618  | 0.555   | 0.593   | 10.145  | 7.230  | 5.993   | 8.689   |
| r30+n0.5 | 0.674   | 0.612  | 0.579   | 0.616   | 9.953   | 7.048  | 6.156   | 8.673   |

TABLE II

Accuracy (higher is better) and Brier score (lower is better), mean over 5 seeds. TS has the same accuracy as MLP by construction.

| Shift    | Acc MLP | Acc TS | Acc MCD | Acc Ens | BRIER MLP | BRIER TS | BRIER MCD | BRIER Ens |
|----------|---------|--------|---------|---------|-----------|----------|-----------|-----------|
| r0       | 0.972   | 0.972  | 0.977   | 0.973   | 0.047     | 0.047    | 0.037     | 0.045     |
| r10      | 0.812   | 0.812  | 0.839   | 0.825   | 0.287     | 0.271    | 0.227     | 0.257     |
| r20      | 0.650   | 0.650  | 0.657   | 0.653   | 0.562     | 0.520    | 0.502     | 0.527     |
| r30      | 0.400   | 0.400  | 0.412   | 0.412   | 1.012     | 0.935    | 0.899     | 0.949     |
| r45      | 0.132   | 0.132  | 0.139   | 0.128   | 1.657     | 1.606    | 1.559     | 1.636     |
| r60      | 0.139   | 0.139  | 0.136   | 0.131   | 1.674     | 1.636    | 1.616     | 1.660     |
| n0.25    | 0.839   | 0.839  | 0.861   | 0.852   | 0.258     | 0.241    | 0.205     | 0.236     |
| n0.5     | 0.496   | 0.496  | 0.516   | 0.500   | 0.867     | 0.812    | 0.749     | 0.813     |
| n0.75    | 0.301   | 0.301  | 0.303   | 0.301   | 1.244     | 1.171    | 1.097     | 1.163     |
| n1.0     | 0.217   | 0.217  | 0.225   | 0.226   | 1.407     | 1.331    | 1.246     | 1.307     |
| r30+n0.5 | 0.200   | 0.200  | 0.196   | 0.203   | 1.425     | 1.345    | 1.303     | 1.354     |

TABLE III

Temperature scaling versus the uncalibrated MLP per shift.  $\Delta\text{ECE}\%$  is the relative change of mean ECE;  $p$  is a paired  $t$ -test over 5 seeds.

| Shift    | ECE MLP | ECE TS | $\Delta\text{ECE}\%$ | $p$   | NLL MLP | NLL TS | $p$   |
|----------|---------|--------|----------------------|-------|---------|--------|-------|
| r0       | 0.018   | 0.016  | -12                  | 0.268 | 0.117   | 0.104  | 0.094 |
| r10      | 0.104   | 0.070  | -33                  | 0.018 | 0.745   | 0.614  | 0.009 |
| r20      | 0.226   | 0.161  | -28                  | 0.017 | 1.954   | 1.450  | 0.012 |
| r30      | 0.457   | 0.388  | -15                  | 0.016 | 4.747   | 3.365  | 0.009 |
| r45      | 0.816   | 0.785  | -4                   | 0.038 | 13.431  | 9.490  | 0.011 |
| r60      | 0.827   | 0.803  | -3                   | 0.099 | 15.266  | 11.101 | 0.012 |
| n0.25    | 0.105   | 0.067  | -36                  | 0.016 | 0.763   | 0.587  | 0.020 |
| n0.5     | 0.392   | 0.335  | -14                  | 0.017 | 4.109   | 2.903  | 0.016 |
| n0.75    | 0.584   | 0.524  | -10                  | 0.012 | 7.746   | 5.468  | 0.011 |
| n1.0     | 0.674   | 0.618  | -8                   | 0.017 | 10.145  | 7.230  | 0.010 |
| r30+n0.5 | 0.674   | 0.612  | -9                   | 0.020 | 9.953   | 7.048  | 0.010 |

Overall picture. Every system is accurate and well calibrated in distribution (ECE below 0.02 for all four in Table I) and becomes severely over-confident under shift: for the MLP at r60, accuracy is 0.139 (Table II) while mean confidence remains close to 1 (Fig. 2). ECE, NLL and Brier all increase monotonically with intensity for each family (Fig. 1). Accuracy is nearly identical across systems (differences of at most a few points, e.g. MCD 0.839 vs MLP 0.812 at r10), so the differences below are mainly calibration, not accuracy.

H1 (TS helps in distribution): not supported by the registered test. At r0, TS lowers mean NLL from 0.117 to 0.104 and ECE from 0.018 to 0.016, but the paired tests give  $p = 0.094$  and  $p = 0.268$  (Table III, Table IV). The direction agrees with the hypothesis; five seeds on a 450-image test set cannot show that the in-distribution gain is real, which is plausible because the MLP is already calibrated here (the fitted  $T$  averages 1.522, Table V).

H2 (ID-fitted TS does not stay calibrated): supported for ECE; the NLL clause is refuted. TS ECE at r60 is 0.803

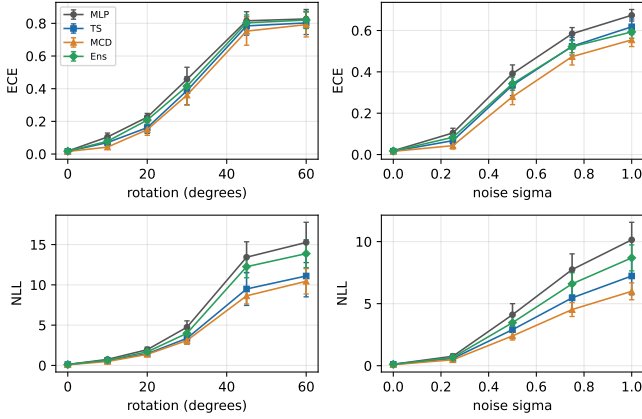


Fig. 1. ECE (top) and NLL (bottom) against shift intensity for rotation (left) and noise (right); mean  $\pm$  SD over 5 seeds. The r0 point is shared by both families.

versus 0.016 at r0 (a ratio of about 50) and 0.618 at n1.0 (about 39), far above the registered factor 2 (Table IV). Even at mild shift ECE rises from 0.016 to 0.070 at r10 and 0.067 at n0.25 (Table III). The relative ECE improvement of TS over the MLP shrinks with rotation, from  $-33\%$  at r10 to  $-3\%$  at r60 ( $p = 0.099$ ), so at the strongest rotation TS is indistinguishable from no calibration; for noise it is  $-36\%$  at n0.25 and  $-8\%$  at n1.0. The second part of H2 as registered (NLL gain over the MLP shrinks with intensity) is wrong: TS lowers NLL at every shifted condition ( $p \leq 0.020$ , Table III) and the absolute gain grows (0.117 to 0.104 at r0, 15.27 to 11.10 at r60 as in Table I). TS therefore reduces the damage under shift, but does not restore calibration; the reduction is largely a better NLL, not a calibrated one.

H3 (ensemble best on NLL and Brier at severe shift): refuted. At r60 and n1.0 MC dropout has the lowest NLL (10.442 and 5.993) and Brier score (1.616 and 1.246) of the four systems, while the ensemble has NLL 13.874 and 8.689 (Tables I, II, and rows H3 of Table IV). The ensemble improves on the single MLP in NLL and Brier at every shift, but its NLL is higher than that of TS at every condition (r60: 13.874 vs 11.101).

H4 (ensemble ECE below MC dropout at severe shift): refuted. The sign is the opposite: MCD ECE is lower at r60 (0.792 vs 0.820,  $p = 0.071$ ) and n1.0 (0.555 vs 0.593, paired  $p$  rounds to 0.000; Table IV). MC dropout has the lowest mean ECE and NLL in all eleven conditions (paired differences from the ensemble are significant at  $\alpha = 0.05$ , uncorrected, for NLL in all 11, ECE in 9 and Brier in 7; the non-significant ECE cases include r0 and r60), although at r0 its ECE ties with TS to three decimals (Table I). This contradicts the common finding that ensembles are the stronger uncertainty method [2], [6]; at least part of our result is a regularisation effect of the dropout-trained network (Section VI), and the ensemble members here have no regularisation and are only 5 in number.

TABLE IV

Registered hypothesis tests. A and B are the compared quantities; for H3 A is the ensemble and B the best other system; for H2 the last column is the ratio.  $p$ : paired  $t$ -test over 5 seeds.

| Hyp. | Quantity                       | Shift | A      | B      | A-B / ratio, $p$  |
|------|--------------------------------|-------|--------|--------|-------------------|
| H1   | TS-MLP nll                     | r0    | 0.104  | 0.117  | -0.013, $p=0.094$ |
| H1   | TS-MLP ece                     | r0    | 0.016  | 0.018  | -0.002, $p=0.268$ |
| H2   | TS ECE shifted vs rot0 (ratio) | r60   | 0.803  | 0.016  | 50.245            |
| H2   | TS ECE shifted vs rot0 (ratio) | n1.0  | 0.618  | 0.016  | 38.689            |
| H3   | nll: best system               | r60   | 13.874 | 10.442 | 3.432             |
| H3   | brier: best system             | r60   | 1.660  | 1.616  | 0.044             |
| H3   | nll: best system               | n1.0  | 8.689  | 5.993  | 2.696             |
| H3   | brier: best system             | n1.0  | 1.307  | 1.246  | 0.061             |
| H4   | Ens-MCD ece                    | r60   | 0.820  | 0.792  | 0.028, $p=0.071$  |
| H4   | Ens-MCD ece                    | n1.0  | 0.593  | 0.555  | 0.039, $p=0.000$  |
| H5   | Oracle-TS ece                  | r45   | 0.053  | 0.785  | -0.732, $p=0.000$ |
| H5   | Oracle-TS ece                  | r60   | 0.029  | 0.803  | -0.773, $p=0.000$ |
| H5   | Oracle-TS ece                  | n0.75 | 0.067  | 0.524  | -0.457, $p=0.000$ |
| H5   | Oracle-TS ece                  | n1.0  | 0.045  | 0.618  | -0.573, $p=0.000$ |

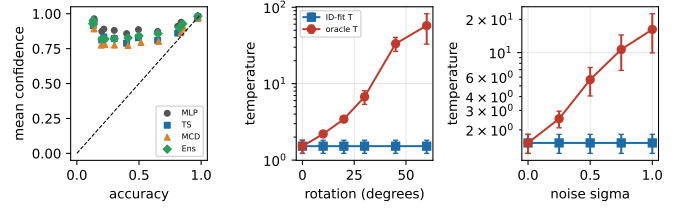


Fig. 2. Left: mean confidence against accuracy for each system and each of the 11 conditions (points on the dashed line are calibrated on average). Middle, right: ID-fitted and oracle temperature (mean  $\pm$  SD over 5 seeds) for rotation and noise.

Where the method fails. (i) At r45 and r60 all four systems are at chance accuracy and ECE is 0.75–0.83: no system is usable and scalar rescaling cannot help (Table I). (ii) The combined shift r30+n0.5 gives ECE of 0.612 for TS, higher than either r30 (0.388) or n0.5 (0.335) alone. (iii) ID-fitted TS is never the best system on ECE or NLL outside r0 and is never well calibrated outside r0.

## VI. Ablations and Analysis

Oracle temperature (tests H5): supported, with a caveat. Fitting  $T$  on shift-matched validation data lowers ECE relative to the ID-fitted  $T$  at every severe condition tested (r45, r60, n0.75, n1.0), e.g. r60 0.029 versus 0.803 (paired  $p$  rounds to 0.000 in each case, Table IV), and over all eleven conditions the oracle ECE stays at or below 0.092 (Table V). The fitted oracle temperature grows with intensity, from 1.522 at r0 to 57.595 at r60 (Fig. 2, Table V), whereas the ID-fitted  $T$  stays at 1.522. So the miscalibration of ID-fitted TS under shift is a temperature mismatch rather than a limit of scalar rescaling for ECE. The caveat is in the NLL columns: the oracle NLL at r60 is 2.279, close to the  $\ln 10$  of a uniform prediction over ten classes, i.e. the oracle “calibrates” by making predictions nearly uniform. That is calibrated but uninformative; accuracy is unchanged. The oracle also needs to know the shift, which is not available in deployment, and the very large temperatures are fitted on only 269 validation images. We cannot say from this

TABLE V

Oracle ablation: temperature fitted on in-distribution validation data (TS) versus on validation data shifted like the test condition (Orc). The ID-fitted  $T$  is constant across shifts by construction.

| Shift    | ECE TS | ECE Orc | NLL TS | NLL Orc | $T$ TS | $T$ Orc |
|----------|--------|---------|--------|---------|--------|---------|
| r0       | 0.016  | 0.016   | 0.104  | 0.104   | 1.522  | 1.522   |
| r10      | 0.070  | 0.056   | 0.614  | 0.561   | 1.522  | 2.197   |
| r20      | 0.161  | 0.076   | 1.450  | 1.103   | 1.522  | 3.438   |
| r30      | 0.388  | 0.092   | 3.365  | 1.765   | 1.522  | 6.737   |
| r45      | 0.785  | 0.053   | 9.490  | 2.249   | 1.522  | 33.365  |
| r60      | 0.803  | 0.029   | 11.101 | 2.279   | 1.522  | 57.595  |
| n0.25    | 0.067  | 0.043   | 0.587  | 0.504   | 1.522  | 2.520   |
| n0.5     | 0.335  | 0.067   | 2.903  | 1.511   | 1.522  | 5.692   |
| n0.75    | 0.524  | 0.067   | 5.468  | 1.974   | 1.522  | 10.676  |
| n1.0     | 0.618  | 0.045   | 7.230  | 2.136   | 1.522  | 16.211  |
| r30+n0.5 | 0.612  | 0.033   | 7.048  | 2.189   | 1.522  | 18.446  |

TABLE VI

Dropout decomposition (exploratory): MLP, the dropout-trained network evaluated deterministically with dropout off (Det), and MC dropout with 20 passes (MCD). Means over 5 seeds.

| Shift    | Acc MLP | Acc Det | Acc MCD | ECE MLP | ECE Det | ECE MCD | NLL Det | NLL MCD |
|----------|---------|---------|---------|---------|---------|---------|---------|---------|
| r0       | 0.972   | 0.976   | 0.977   | 0.018   | 0.018   | 0.016   | 0.084   | 0.083   |
| r10      | 0.812   | 0.840   | 0.839   | 0.104   | 0.070   | 0.042   | 0.538   | 0.491   |
| r20      | 0.650   | 0.653   | 0.657   | 0.226   | 0.203   | 0.149   | 1.594   | 1.370   |
| r30      | 0.400   | 0.412   | 0.412   | 0.457   | 0.427   | 0.361   | 3.732   | 3.098   |
| r45      | 0.132   | 0.139   | 0.139   | 0.816   | 0.787   | 0.752   | 10.575  | 8.639   |
| r60      | 0.139   | 0.135   | 0.136   | 0.827   | 0.820   | 0.792   | 12.702  | 10.442  |
| n0.25    | 0.839   | 0.858   | 0.861   | 0.105   | 0.080   | 0.043   | 0.556   | 0.479   |
| n0.5     | 0.496   | 0.512   | 0.516   | 0.392   | 0.360   | 0.280   | 3.059   | 2.402   |
| n0.75    | 0.301   | 0.310   | 0.303   | 0.584   | 0.553   | 0.474   | 5.869   | 4.525   |
| n1.0     | 0.217   | 0.227   | 0.225   | 0.674   | 0.643   | 0.555   | 7.791   | 5.993   |
| r30+n0.5 | 0.200   | 0.198   | 0.196   | 0.674   | 0.654   | 0.579   | 7.707   | 6.156   |

TABLE VII

Sensitivity sweeps (mean over seeds 0–2, conditions r0, r30, n0.5): ensemble size  $M$  and dropout rate  $p$ . The  $M = 5$  and  $p = 0.2$  rows are deliberate re-runs of the main configuration on 3 of the 5 seeds.

| Sweep | Value | ECE r0 | NLL r0 | ECE r30 | NLL r30 | ECE n0.5 | NLL n0.5 |
|-------|-------|--------|--------|---------|---------|----------|----------|
| M     | 1     | 0.016  | 0.098  | 0.452   | 4.560   | 0.404    | 4.367    |
| M     | 2     | 0.017  | 0.095  | 0.452   | 4.401   | 0.370    | 3.984    |
| M     | 3     | 0.015  | 0.094  | 0.424   | 3.978   | 0.354    | 3.715    |
| M     | 5     | 0.015  | 0.092  | 0.419   | 3.853   | 0.353    | 3.589    |
| M     | 10    | 0.014  | 0.091  | 0.401   | 3.726   | 0.345    | 3.449    |
| $p$   | 0.1   | 0.014  | 0.078  | 0.417   | 3.524   | 0.343    | 3.088    |
| $p$   | 0.2   | 0.015  | 0.071  | 0.364   | 3.045   | 0.295    | 2.515    |
| $p$   | 0.3   | 0.021  | 0.073  | 0.336   | 2.721   | 0.260    | 2.218    |
| $p$   | 0.5   | 0.038  | 0.095  | 0.245   | 2.178   | 0.194    | 1.760    |

study whether a practical shift-agnostic procedure, such as perturbing the calibration set [4], would find them.

Why does MC dropout win? A decomposition (exploratory). The dropout-trained network evaluated deterministically (Det) is not identical to the MLP: its ECE is between MLP and MCD at most conditions (e.g. r30 0.457, 0.427, 0.361; n1.0 0.674, 0.643, 0.555; Table VI), while at r0 the three are within 0.003 of each other. Accuracy of Det is higher than the MLP at r10 (0.840 vs 0.812) and n0.25 (0.858 vs 0.839). The paired Det-versus-MLP ECE differences are not significant at r0 ( $p = 0.95$ ), r30 ( $p = 0.086$ ) or r60 ( $p = 0.37$ ) and only at n1.0 ( $p = 0.004$ ),

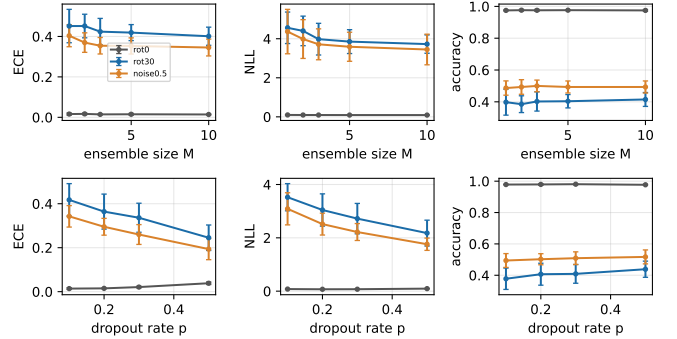


Fig. 3. Sweeps over ensemble size (top) and dropout rate (bottom); mean  $\pm$  SD over seeds 0–2.

whereas Det versus MCD is significant at r30, r60 and n1.0 ( $p \leq 0.023$ ) but not at r0 ( $p = 0.47$ ). So the evidence that MC averaging at test time contributes is clearer than that for dropout regularisation, which may also contribute, and a part of MC dropout’s advantage over the ensemble is a property of the differently trained network rather than of its test-time Bayesian approximation (cf. [7]). We did not test a dropout-regularised ensemble, which would be the natural controlled comparison.

Sensitivity to ensemble size and dropout rate. In Table VII the ensemble’s NLL under r30 falls monotonically from 4.560 ( $M = 1$ ) to 3.726 ( $M = 10$ ), and under n0.5 from 4.367 to 3.449; ECE falls from 0.452 to 0.401 (r30) and 0.404 to 0.345 (n0.5). Even  $M = 10$  stays above MC dropout at  $p = 0.2$  on r30 (NLL 3.045, ECE 0.364) and n0.5 (NLL 2.515, ECE 0.295), so the H3/H4 verdicts are not an artefact of  $M = 5$ . Larger dropout rates reduce shifted ECE and NLL further ( $p = 0.5$ : r30 ECE 0.245, n0.5 ECE 0.194) but worsen in-distribution calibration (r0 ECE 0.038 versus 0.015 at  $p = 0.2$ ), a trade-off that the main comparison at  $p = 0.2$  does not reveal. The sweeps use seeds 0–2 only; they are not used to choose  $p$  or  $M$ , and a rate chosen on shifted data would be tuning on the shift.

## VII. Limitations

This is a small CPU study and its scope is narrow. (i) One dataset (8×8 digits), one architecture (a two-hidden-layer MLP) and two synthetic shift families; rotation of 8×8 images is destructive, so accuracy drops to chance at r45 and r60, and the conclusions may not transfer to natural corruptions, CNNs or larger images. (ii) Five seeds (three for the sweeps), a 450-image test set per seed and paired  $t$ -tests without multiple-comparison correction;  $p$ -values are descriptive, and the H1 non-result may be a lack of power. (iii) Baselines are reimplemented, not tuned: no weight decay, early stopping or per-method hyperparameter search, and the MC-dropout network differs from the other systems in training regularisation, which confounds the system comparison (Section VI); the dropout decomposition was exploratory and added after seeing results. (iv) ECE is a binned estimator that is easily dominated by the mean confidence when accuracy is near chance; NLL and Brier are shown alongside but the three metrics are strongly correlated across conditions here. (v) The oracle temperature uses the shift itself and 269 validation images, and achieves low ECE with nearly uniform predictions; it shows a mechanism, not a practical method. (vi) Models are cached per seed so rows across shifts are not independent trainings; shift noise is a single draw per seed. A fuller study would add CNNs and corrupted natural-image benchmarks, regularised and larger ensembles, post-hoc methods designed for shift, more seeds, and a proper OOD-detection evaluation.

## VIII. Conclusion

On digits with rotation and noise shift, a temperature fitted on in-distribution validation data did not stay calibrated: TS ECE rose from 0.016 in distribution to 0.803 at r60 and 0.618 at n1.0 (about 50 and 39 times), although TS still lowered NLL relative to the uncalibrated MLP at all shifted conditions and its ECE gain over the MLP shrank to 3% at r60. A temperature fitted on shift-matched validation data brought ECE to at most 0.092 in all conditions, so the failure is a temperature

mismatch for ECE, but it did so partly by producing near-uniform predictions. Contrary to our pre-registered expectation and to the literature we cite, MC dropout, not the 5-member ensemble, had the lowest ECE, NLL and Brier in this setting, in part (per an exploratory decomposition) because of the regularisation of the dropout-trained network. The in-distribution benefit of TS was not statistically supported at 5 seeds. These findings are limited to a small MLP on a small dataset and a handful of seeds.

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